The Equation Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: SOLVE, DISTRIB, LITERAL and MODEL (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** SOLVE
- ☐ **A2** SOLVE
- ☐ **A3** DISTRIB
- ☐ **A4** DISTRIB
- ☐ **A5** LITERAL
- ☐ **A6** MODEL

Entrées

10 pts each · choose 4

- ☐ **E1** SOLVE
- ☐ **E2** DISTRIB
- ☐ **E3** LITERAL
- ☐ **E4** LITERAL
- ☐ **E5** MODEL
- ☐ **E6** SOLVE

Chef's Specials

20 pts each · choose 2

- ☐ **S1** SOLVE
- ☐ **S2** DISTRIB
- ☐ **S3** LITERAL
- ☐ **S4** MODEL

My skill check (circle each skill you have covered): SOLVE · DISTRIB · LITERAL · MODEL

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 SOLVE ☐ My pick · 5 pts

Solve: 3x − 7 = 11

A2 SOLVE ☐ My pick · 5 pts

Solve: 5x + 4 = 2x + 19

A3 DISTRIB ☐ My pick · 5 pts

Simplify: 4(2x − 3) + 5x

A4 DISTRIB ☐ My pick · 5 pts

Solve: 3(x + 4) = 21

A5 LITERAL ☐ My pick · 5 pts

Solve for *y*: 2x + y = 9

A6 MODEL ☐ My pick · 5 pts

A plumber charges a $40 visit fee plus $25 per hour. Write an equation for a $140 bill and solve it for the hours *h*.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 SOLVE ☐ My pick · 10 pts

Solve: $4(2x - 1) - 3 = 5(x + 2) + 1$

E2 DISTRIB ☐ My pick · 10 pts

Simplify $3(2x - 5) - 2(x - 4) + 7$ completely, then evaluate your result when $x = -3$.

E3 LITERAL ☐ My pick · 10 pts

The area of a trapezoid is $A = \frac{h(b + c)}{2}$.

(a) Solve the formula for b.

(b) Find b when $A = 42$, $h = 6$, and $c = 5$.

E4 LITERAL ☐ My pick · 10 pts

Solve for x: $3(x - a) = 2x + b$.

Then find x when $a = 4$ and $b = -5$.

E5 MODEL ☐ My pick · 10 pts

Gym A charges a \$20 joining fee plus \$15 per month. Gym B charges \$50 plus \$10 per month. Write and solve an equation for the number of months m when the total costs are equal. What is that total cost?

E6 SOLVE ☐ My pick · 10 pts

Dev solved $2(3x - 4) = 4x + 6$ like this:

$6x - 4 = 4x + 6$

$2x = 10$

$x = 5$

Find his mistake, explain it, and give the correct solution.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 SOLVE ☐ My pick · 20 pts

Mia thinks of a number n, multiplies it by 4, subtracts 6, divides the result by 2, and then adds 5. She gets 20.

(a) Write an equation for her steps.

(b) Solve it to find n.

(c) Check by following her steps forward.

S2 DISTRIB ☐ My pick · 20 pts

Lena solved $3(x - 2) - 2(x - 5) = 4x + 1$:

$3x - 6 - 2x - 10 = 4x + 1$

$x - 16 = 4x + 1$

$-3x = 17,\ x = -\frac{17}{3}$

(a) Find and explain her mistake. **(b)** Solve correctly. **(c)** Check in the original equation.

S3 LITERAL ☐ My pick · 20 pts

The balance of an account with simple interest is $A = P + Prt$.

(a) Solve for t.

(b) Solve for P. (Hint: factor first.)

(c) Find P when $A = 1{,}260$, $r = 0.05$, and $t = 4$.

S4 MODEL ☐ My pick · 20 pts

A rectangle’s length is 4 cm more than 3 times its width. Its perimeter is 72 cm.

(a) Define a variable and write an equation.

(b) Find the length and width.

(c) A square has the same perimeter. Which shape has the greater area, and by how much?